

Zhen Qian

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RESEARCH EXPERIENCES

Potsdam Institute for Climate Impact Research <i>Guest researcher</i>	03/2024 – Present <i>Potsdam, Germany</i>
Nanjing Normal University <i>Research Assistant</i>	07/2023 – 02/2024 <i>Nanjing, China</i>

EDUCATION

Technical University of Munich <i>Doctoral student in Earth System Modeling</i> <i>Supervisor: Prof. Dr. Niklas Boers</i>	03/2024 – Present <i>Munich, Germany</i>
Nanjing Normal University <i>M.S in GIScience (awarded National Scholarship)</i> <i>Supervisor: Prof. Dr. Min Chen</i>	09/2020 – 06/2023 <i>Nanjing, China</i>
Nantong University <i>B.S in GIScience (awarded National Scholarship)</i>	09/2016 – 06/2020 <i>Nantong, China</i>

RESEARCH INTERESTS

Human-Earth System Observation and Modeling

Forest ecosystem; Urban environment; Climate change

Geoinformatics and Remote Sensing

Space-time data curation, processing, analysis, modeling and visualization

Machine Learning and Deep Learning

AI for Earth Science and Sustainability; Multi-modality; Explainable AI; Generative AI

RESEARCH OUTPUTS

Publication statistics

- Over 20 peer-reviewed publications (10 as first author, including *Nature Reviews Earth & Environment*, *Nature Climate change*, *Nature Geoscience*, *Earth System Science Data*, *Sustainable Cities and Society*)
- Citations: 973; h-index: 15; i10-index: 16 (statistics by [Google Scholar](#))

Recent featured publications

(Note: “†” indicates equal contribution; complete publication list available at [Google Scholar](#))

1. *Worldwide rooftop photovoltaic electricity generation may mitigate global warming*
Zhang, Z.†, **Qian, Z.†**, Chen, M., Zhu, R., Zhang, F., Zhong, T., Lin, J., Ning, L., Xie, W., Creutzig, F., Tang, W., Liu, L., Yang, J., Pu, Y., Cai, W., Pu, Y., Liu, D., Yang, H., Su, H., Lu, M., Li, F., Cui, X., Xie, Z., Sheng, T., Zhang, K., Santi, P., Tian, L., Lü, G. and Yan, J.
Nature Climate Change, 15, 393–402, 2025 (IF 29.6)

2. *Collaboration between artificial intelligence and Earth science communities for mutual benefit*
Chen, M. †, **Qian, Z.** †, Boers, N., Creutzig, F., Camps-Valls, G., Hubacek, K., Claramunt, C., Wilson, J. P., Nativi, S., Jakeman, A. J., Müller, R. D., Batty, M., Zhou, C., Chen, F., Wang, Q., Zhang, F., Barton, C. M., Strobl, J., Meadows, M., Ratti, C., Hess, P., Xu, Q., Zhang, Z., Gu, Q., Zhu, A.-X., Lin, H., Yuan, L. and Lü, G.
Nature Geoscience, 17, 949–952, 2024 (IF 15.7)
3. *Simultaneous extraction of diverse building information across large-scale urban landscapes using high-resolution satellite imagery*
Qian, Z., Chen, M., Sun, Z., Zhang, F., Xu, Q., Guo, J., Xie, Z. and Zhang, Z.
Sustainable Cities and Society, 106, 105393, 2024 (IF 10.5)
4. *Iterative integration of deep learning in hybrid Earth surface system modelling*
Chen, M. †, **Qian, Z.** †, Boers, N., Jakeman, A. J., Kettner, A. J., Brandt, M., Kwan, M.-P., Batty, M., Li, W., Zhu, R., Luo, W., Ames, D. P., Barton, C. M., Cuddy, S. M., Koirala, S., Zhang, F., Ratti, C., Liu, J., Zhong, T., Liu, J., Wen, Y., Yue, S., Zhu, Z., Zhang, Z., Sun, Z., Lin, J., Ma, Z., He, Y., Xu, K., Zhang, C., Lin, H. and Lü, G.
Nature Reviews Earth & Environment, 4, 568–581, 2023 (IF 49.7)
5. *Vectorized dataset of roadside noise barriers in China using street view imagery*
Qian, Z., Chen, M., Yang, Y., Zhong, T., Zhang, F., Zhu, R., Zhang, K., Zhang, Z., Sun, Z., Ma, P., Lü, G. and Yan, J.
Earth System Science Data, 14, 4057–4076, 2022 (IF 11.2)
6. *Deep Roof Refiner: A detail-oriented deep learning network for refined delineation of roof structure lines using satellite imagery*
Qian, Z., Chen, M., Zhong, T., Zhang, F., Zhu, R., Zhang, Z., Zhang, K., Sun, Z. and Lü, G.
International Journal of Applied Earth Observation and Geoinformation, 107, 102680, 2022 (IF 7.6)
7. *Identification of urban functional areas by coupling satellite images and taxi GPS trajectories*
Qian, Z., Liu, X., Tao, F. and Zhou, T.
Remote Sensing, 12, 2449, 2020 (IF 5.0)

Working papers

1. *Decadal sink-source shifts of forest aboveground carbon since 1988* (Submitted)
Qian, Z., Bathiany, S., Liu, T., Blaschke, LL., Teo, H. and Boers, N.

SELECTED AWARDS & HONORS

Awards

Top 10 Research Achievements of Chinese Geography	2024
Top 10 Sci-tech Achievements of China Society of Territorial Economists	2023
Provincial Excellent Master's Thesis	2023
Provincial Outstanding Graduates	2023
Pacemaker to Merit Student in University for Postgraduates	2022
Excellent Student Cadre in University for Postgraduates	2021
Provincial Excellent Undergraduate Thesis	2021

Provincial Excellent Student Cadre for Undergraduates	2019
Pacemaker to Merit Student in University for Undergraduates	2018
Excellent Student Cadre in University for Undergraduates	2017, 2019

Scholarships

President's Scholarship for Postgraduates	2023
<i>Ranked Top 0.1% / Award amount: 10k ¥</i>	
National Scholarship for Postgraduates	2022
<i>Ranked Top 0.1% / Award amount: 20k ¥</i>	
First Prize Scholarship in University for Postgraduates	2020, 2021, 2022
<i>Ranked Top 1% / Award amount: 3×12k ¥</i>	
National Scholarship for Undergraduates	2018
<i>Ranked Top 0.1% / Award amount: 8k ¥</i>	
Enterprise Scholarship for Undergraduates	2017, 2019
<i>Ranked Top 1% / Award amount: 2×2.5k ¥</i>	
First Prize Scholarship in University for Undergraduates	2017, 2018, 2019
<i>Ranked Top 1% / Award amount: 3×1.5k ¥</i>	

RESEARCH GRANTS

Provincial Research and Practice Innovation Program	2022-2023
<i>Title: Spatial intelligent analysis considering geographic priors; PI</i>	<i>Funding amount: 15k ¥</i>
Provincial Research and Practice Innovation Program	2018-2019
<i>Title: Travel behavior detection based on artificial intelligence; PI</i>	<i>Funding amount: 8k ¥</i>

ACADEMIC ACTIVITIES

Reviewer Services

All Earth; Cities; Computers, Environment and Urban Systems; Earth Science Informatics; Geo-spatial Information Science; IEEE International Geoscience and Remote Sensing (IGARSS) 2025; IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing; International Journal of Applied Earth Observation and Geoinformation; Journal of Environmental Management; Scientific Reports; The Journal of Supercomputing; Urban Climate

Presentations

The Ninth Conference on Virtual Geographic Environment	08/2023
<i>Title: Simultaneous extraction of building information across large-scale landscapes</i>	<i>Yanji</i>
The Eighth Conference on Virtual Geographic Environment	07/2022
<i>Title: Roof structural line delineation based on deep learning and satellite imagery</i>	<i>Ganzhou</i>

SPECIALIZED SKILLS

Languages

Mandarin (Native), English (Professional proficiency), German (Elementary)

Programming

Python (Proficient, including OpenCV, Pytorch, Numpy, Pandas, Matplotlib, GDAL, etc.), R, Matlab

Professional tools

ArcGIS, QGIS, GEE, etc